PROJECT REPORT OF CO-OP PROJECT AT INDUSTRY (MODULE-2)

ON

YOUTH MENTAL WELLNESS PLATFORM USING FINE-TUNED BERT FOR SUICIDAL IDEATION DETECTION

submitted in partial fulfilment of the requirements for the award of degree of

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Abstract

The increasing prevalence of mental health disorders among youth, combined with barriers such as social stigma, privacy concerns, and limited access to professional care, has created an urgent need for scalable and accessible mental health solutions. This project presents the design and development of a **privacy-first Youth Mental Wellness Platform** powered by advanced Natural Language Processing (NLP) techniques.

The core innovation of the system lies in leveraging a fine-tuned transformer-based model, specifically **BERT (Bidirectional Encoder Representations from Transformers)**, to detect suicidal ideation and high-risk language patterns in textual inputs with high accuracy and recall. Unlike traditional rule-based chatbots, the platform integrates an empathetic AI companion, AI-driven journaling, and explainable mood analytics to provide meaningful user engagement.

The system incorporates a tiered crisis escalation workflow that ensures timely intervention. When high-risk behavior is detected, the system provides immediate empathetic responses, suggests relevant mental health resources, and enables optional human-in-the-loop moderation. Privacy is prioritized through data anonymization and minimal data storage policies.

Experimental evaluation demonstrates that the fine-tuned BERT model achieves an F1-score of 0.98, outperforming traditional machine learning models such as Logistic Regression, Naive Bayes, Random Forest, and Linear SVC. This highlights the effectiveness of contextual deep learning models in understanding nuanced human language.

Overall, the platform aims to provide an intelligent, scalable, and ethical solution for early detection and intervention in youth mental health, bridging the gap between vulnerable individuals and timely support systems.

Methodology / Approach

The proposed system follows a **privacy-first AI-driven architecture** designed as a Progressive Web Application (PWA). The core objective is to process user-generated textual data and identify potential mental health risks using advanced NLP techniques.

System Overview

The system consists of an NLP pipeline that processes user inputs from chat interactions and journaling features. The pipeline includes:

- Data ingestion and sanitization
- Text preprocessing and tokenization
- Transformer-based classification
- Risk assessment and response generation

Transformer-Based Model

The platform utilizes a fine-tuned **BERT model**, which leverages the Transformer architecture and self-attention mechanisms to understand contextual relationships in text. Unlike traditional models, BERT processes entire sentences simultaneously, allowing it to capture nuanced emotional expressions.

Model Fine-Tuning

The pre-trained `bert-base-uncased` model is fine-tuned for binary classification (Suicidal vs Non-Suicidal). A classification head consisting of a dropout layer and linear layer is added. The model is trained using cross-entropy loss with class-weighting to handle imbalanced datasets.

Crisis Escalation Workflow

The system implements a tiered intervention strategy:

1. **Tier 1: Immediate Support** – Empathetic AI responses for high-risk inputs
2. **Tier 2: Resource Recommendation** – Suggests helplines and support services
3. **Tier 3: Human Intervention** – Optional escalation to human moderators

Tools and Technology

Machine Learning:

- **BERT (bert-base-uncased)** – A transformer-based deep learning model used for contextual text understanding. It leverages bidirectional encoding and self-attention mechanisms to capture semantic relationships within sentences, enabling accurate detection of nuanced expressions such as suicidal ideation, sarcasm, and negation.
- **PyTorch / TensorFlow** – Deep learning frameworks used for model training, fine-tuning, and deployment. These frameworks provide GPU acceleration, efficient tensor operations, and support for transformer architectures, enabling scalable and high-performance model execution.
- **Scikit-learn** – Used for implementing and evaluating baseline machine learning models such as Logistic Regression, Support Vector Machines (SVM), Naive Bayes, and Random Forest. It also provides tools for data preprocessing, model evaluation, and performance comparison using metrics like precision, recall, and F1-score.

Other Tools:

- **NLP Preprocessing Libraries (NLTK / SpaCy / HuggingFace Tokenizers)** – Used for text cleaning, tokenization, stop-word removal, and normalization. These libraries ensure that raw textual data is transformed into a structured format suitable for model input.
- **Kaggle Dataset (Suicide and Depression Detection)** – A large-scale dataset consisting of approximately 232,000 text samples collected from moderated online forums. It provides labeled data for training and evaluating classification models, ensuring robustness and generalization.
- **HuggingFace Transformers Library** – Provides pre-trained BERT models and utilities for fine-tuning, tokenization, and model deployment. It simplifies the implementation of transformer-based architectures and accelerates experimentation.
- **Data Visualization Tools (Matplotlib / Seaborn)** – Used to generate performance graphs such as confusion matrices, accuracy plots, and model comparison charts for analysis and presentation.

Experimental Results and Analysis

Dataset and Experimental Setup: The model was trained and evaluated using the Kaggle Suicide and Depression Detection dataset, consisting of approximately 232,000 text samples collected from moderated online forums. The dataset was split into 80% training and 20% testing sets. Preprocessing steps included removal of URLs, usernames, and special characters, followed by tokenization using the BERT tokenizer.

Baseline Models: To evaluate the effectiveness of the proposed approach, the fine-tuned BERT model was compared against traditional machine learning algorithms:

- Naive Bayes (NB)
- Random Forest (RF)
- Logistic Regression (LR)
- Linear Support Vector Classifier (Linear SVC)

Performance Metrics: The models were evaluated using standard classification metrics:

- Precision
- Recall
- F1-Score
- Accuracy

Recall was given higher importance due to the critical need to minimize false negatives in suicidal ideation detection.

Results Analysis: The fine-tuned BERT model significantly outperformed traditional machine learning models, achieving an F1-score of 0.98 and accuracy of approximately 97.93%. In comparison, the best-performing baseline model (Linear SVC) achieved an F1-score of 0.94.

The performance improvement can be attributed to BERT's ability to capture contextual and semantic relationships within text, which traditional models fail to interpret effectively. This is particularly important in distinguishing nuanced expressions such as sarcasm or negation (e.g., "I'm not going to harm myself").

Furthermore, the confusion matrix demonstrates a substantial reduction in false negatives, which is critical in high-risk applications like mental health monitoring.

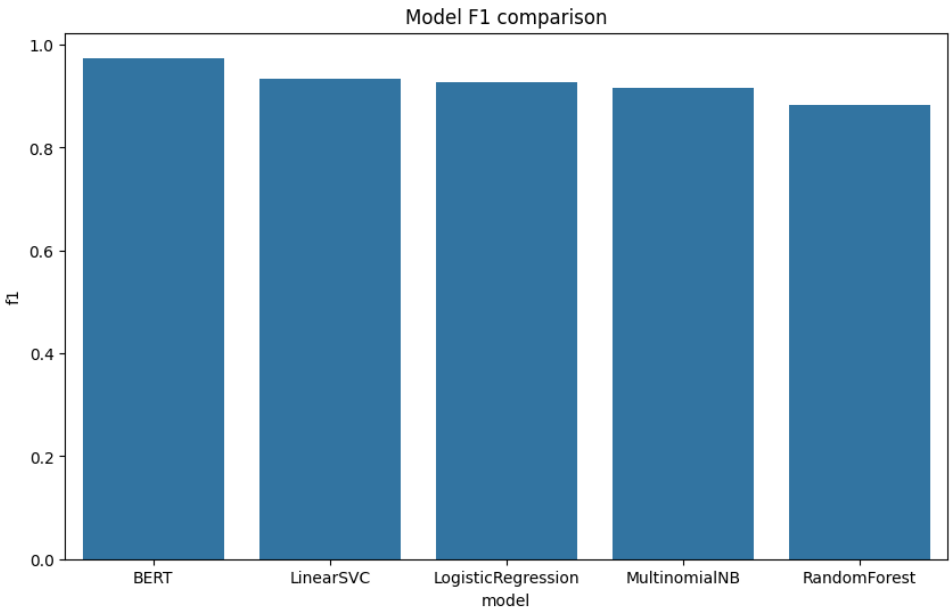

Figure 1: Comparison of Model Performance (Accuracy / F1-Score)

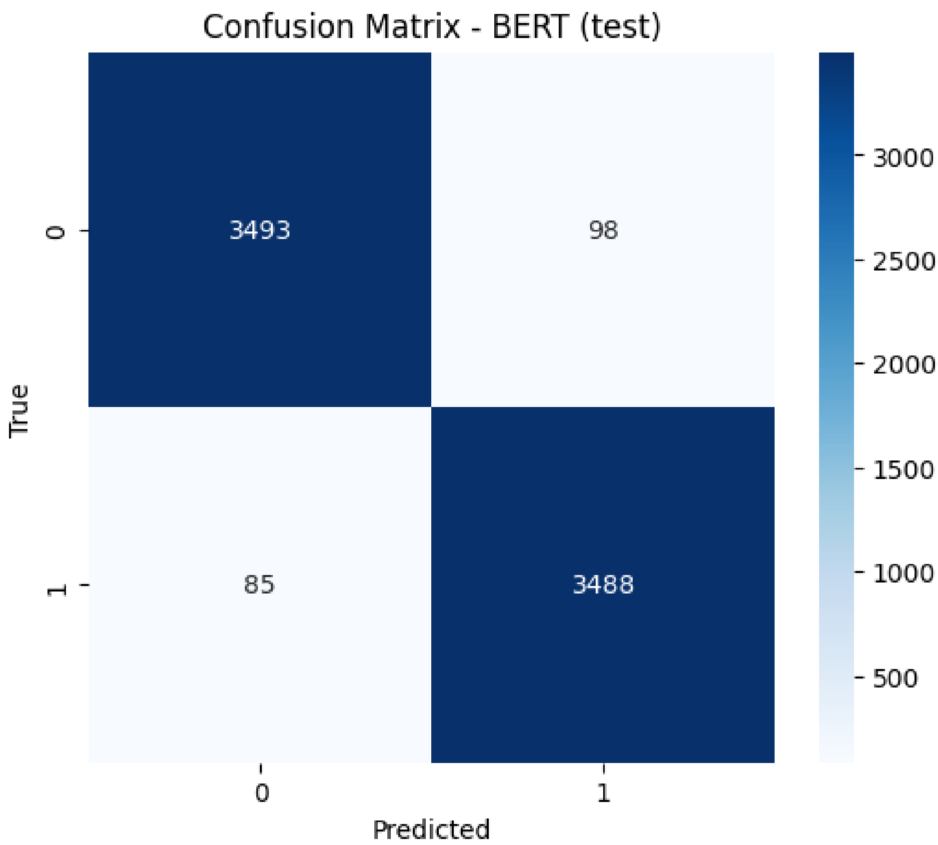

Figure 2: Confusion Matrix of Fine-Tuned BERT Model

Key Insight: Even a small improvement in accuracy leads to a significant reduction in misclassification rates, enhancing the reliability of early risk detection systems.

Expected Outcomes

The system is expected to deliver:

- **Early Detection:** Accurate identification of suicidal ideation using contextual NLP models
- **High Accuracy:** Achieves an F1-score of approximately 0.98, outperforming traditional models
- **Privacy Protection:** Secure handling of sensitive user data with anonymization
- **Real-Time Support:** Immediate AI-generated empathetic responses
- **Crisis Intervention:** Structured escalation to resources and human support
- **User Engagement:** AI-driven journaling and mood tracking features
- **Scalable Solution:** Capable of supporting large user bases with minimal infrastructure

Project Plan

The project timeline is spread over 12 weeks as follows:

Weeks	**Activities**
W1–W2	Requirement analysis, literature review, dataset exploration
W3–W4	Data preprocessing and baseline model implementation
W5–W6	BERT model fine-tuning and evaluation
W7	Backend API development and integration
W8	Frontend development and UI design
W9	Crisis escalation workflow implementation
W10	Privacy mechanisms and optimization
W11	Testing, validation, and performance tuning
W12	Deployment and documentation